

IFN647 Week 2 Review Questions

Question 1 (Text statistic)

Please open a text or an XML file (e.g., “6146.xml”) and represent it as a list of paragraphs or sentences, *text*. You may remove any non-relevant information (e.g., ‘<p>’, ‘</p>’, ‘\n’). After that, you need to find all terms and their frequencies (the number of occurrences in the file) if their length > 2, represent them using a dictionary, *doc*; and print the number of total terms (e.g., 137). Then print the top-10 terms in *doc* in a descending order, e.g.,

[(‘the’, 8), (‘technical’, 2), (‘bounce’, 2), (‘said’, 2), (‘and’, 2), (‘not’, 2), (‘due’, 2), (‘rose’, 2), (‘argentina’s’, 2), (‘argentine’, 1)]

At last, please plot the distribution of the top-10 terms (see, Fig 1 for an example).

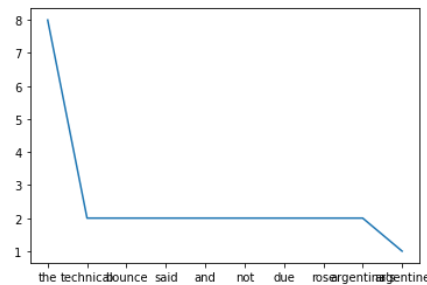


Fig 1. Example of distribution of top-k frequent words.

Question 2 Which of the following is FALSE? And explain why it is FALSE.

- (1) Stemming is a component of text processing that captures the relationships between different variations of a word.
- (2) Stemming reduces the different forms of a word that occur because of inflection (e.g., plurals, tenses) or derivation (e.g., making a verb into a noun by adding the suffixation) to a common stem.
- (3) In general, using a stemmer for search applications with English text produces a small but noticeable improvement in the quality of results.
- (4) A dictionary-based stemmer uses a small program to decide whether two words are related, usually based on knowledge of word suffixes for a particular language.

Question 3 (N-grams)

Typically, n -grams are formed from overlapping sequences of words., i.e. move n -word “window” one word at a time in a document. For example, *bigrams* are 2 words sequences, and *trigrams* are 3 words sequences.

The definition of Tropical fish is described in the following document:

Tropical fish are generally those **fish** found in aquatic **tropical** environments around the world, including both **freshwater** and **saltwater** species. Fishkeepers often keep **tropical fish** in **freshwater** and **saltwater aquariums**.

Please design a python program to print all bigrams and trigrams of the above document that contain at least one of the highlighted key words (‘fish’, ‘tropical’, ‘freshwater’, ‘saltwater’, ‘aquariums’).

Question 4 (Html tags)

- (a) Design a python program to extract all hyperlinks (or destination links) in a html file. You can use HTMLParser python package. You may download a real html file (e.g., “html_example.html”) to test it.
- (b) What ethical issues might exist in extracting hyperlinks from the Web?